

csharp

```
# Generiká
```

```
(1,3): error CS1024: Preprocessor directive expected
```

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Dôležitý koncept, nielen v C#, ale v celom .NET-e i iných jazykov.

Umožňuje:

- zvýšenie výkonu
- typovú bezpečnosť
- viacnásobné použitie raz napísaného kódu

```
(1,19): error CS1044: Cannot use more than one type in a for, using, fixed, or  
declaration statement
```

```
(1,26): error CS1002: ; expected
```

```
(1,29): error CS1040: Preprocessor directives must appear as the first non-  
whitespace character on a line
```

```
(2,1): error CS1002: ; expected
```

```
(3,12): error CS1002: ; expected
```

```
(4,10): error CS1002: ; expected
```

```
(5,15): error CS1002: ; expected
```

```
(5,28): error CS1002: ; expected
```

```
(5,43): error CS1002: ; expected
```

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```
## Test výkonu
```

```
(1,1): error CS1024: Preprocessor directive expected
```

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Jedna z najdôležitejších výhod generík je ****výkon**** (rýchlosť spracovania dát)
Prerekvizita:

```
(1,9): error CS1002: ; expected
(1,32): error CS1002: ; expected
(1,43): error CS1002: ; expected
(1,63): error CS1026: ) expected
(1,63): error CS1002: ; expected
(1,78): error CS1003: Syntax error, ',' expected
(1,81): error CS1002: ; expected
(1,92): error CS1002: ; expected
(1,104): error CS1002: ; expected
(1,124): error CS1002: ; expected
(1,137): error CS1056: Unexpected character '''
(1,138): error CS1002: ; expected
(1,147): error CS1002: ; expected
(1,147): error CS1056: Unexpected character '''
(1,155): error CS1002: ; expected
(1,173): error CS1056: Unexpected character '''
(1,174): error CS1002: ; expected
(1,181): error CS1525: Invalid expression term ''
(1,181): error CS1002: ; expected
(1,181): error CS1056: Unexpected character '''
(1,182): error CS7017: Member definition, statement, or end-of-file expected
(2,14): error CS1733: Expected expression
```

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```
using System.Diagnostics;
var stopwatch = new Stopwatch();

// Počet iterácií (aby boli výsledky hodnoverné)
const int IterationCount = 20;
// Počet čísiel, ktoré sa pridajú do zoznamu a prečítajú zo zoznamu (negeric
const int MaxNumberCount = 10_000_000;

var boxingTimes = new List<double>();
var unboxingTimes = new List<double>();
```

Test rýchlosti negenerickej triedy `ArrayList`:

```
(1,16): error CS1002: ; expected
(1,36): error CS1056: Unexpected character ``
(1,37): error CS1002: ; expected
(1,46): error CS1002: ; expected
(1,46): error CS1056: Unexpected character ``
(1,47): error CS7017: Member definition, statement, or end-of-file expected
```

```
using System.Collections;
using System.Text;
using System.Diagnostics;
using System.Linq;
boxingTimes.Clear();

unboxingTimes.Clear();

for (int c = 0; c < IterationCount; c++)
{
    stopwatch.Restart();
    var list = new ArrayList();
    for (int i = 0; i < MaxNumberCount; i++)
    {
        list.Add(i); // Tu dochádza k boxingu - konverzia hodnotového typu (int)
    }
    boxingTimes.Add(stopwatch.ElapsedMilliseconds);

    stopwatch.Restart();
    var result = new StringBuilder(MaxNumberCount);
    for (int i = 0; i < MaxNumberCount; i++)
    {
        int number = (int)list[i]; // Tu dochádza k unboxingu - konverzia z re
        result.Append(number);
    }
    unboxingTimes.Add(stopwatch.ElapsedMilliseconds);
}
```

```
Console.WriteLine($"ArrayList - pridanie cisiel trvalo priemerne (boxing): {box}");
Console.WriteLine($"ArrayList - precitanie cisiel trvalo priemerne (unboxing): {unbox}");
```

ArrayList - pridanie cisiel trvalo priemerne (boxing): 434.65

ArrayList - precitanie cisiel trvalo priemerne (unboxing): 160.1

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Test rýchlosti generickej triedy `List<int>`:

```
(1,16): error CS1002: ; expected
(1,34): error CS1056: Unexpected character ``
(1,35): error CS1002: ; expected
(1,44): error CS1002: ; expected
(1,44): error CS1056: Unexpected character ``
(1,45): error CS7017: Member definition, statement, or end-of-file expected
```

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```
using System.Collections;
using System.Text;
using System.Diagnostics;
using System.Linq;
boxingTimes.Clear();
unboxingTimes.Clear();

for (int c = 0; c < IterationCount; c++)
{
    stopwatch.Restart();
    var list = new List<int>();
    for (int i = 0; i < MaxNumberCount; i++)
    {
```

```

        list.Add(i); // Tu k boxingu nedochádza
    }
    boxingTimes.Add(stopwatch.ElapsedMilliseconds);

    stopwatch.Restart();
    var result = new StringBuilder(MaxNumberCount);
    for (int i = 0; i < MaxNumberCount; i++)
    {
        int number = list[i]; // Tu k unboxingu nedochádza
        result.Append(number);
    }
    unboxingTimes.Add(stopwatch.ElapsedMilliseconds);
}

Console.WriteLine($"List - pridanie cisel trvalo priemerne: {boxingTimes.Average()} ms");
Console.WriteLine($"List - precitanie cisel trvalo priemerne: {unboxingTimes.Average()} ms");

```

List - pridanie cisel trvalo priemerne: 42.2 ms

List - precitanie cisel trvalo priemerne: 81.2 ms

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Typová bezpečnosť

(1,1): error CS1024: Preprocessor directive expected

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Do negenerického ArrayListu môže byť pridaný akýkoľvek objekt iného typu ako po
 Ak by sme chceli prejsť všetky položky foreach-om, a skonvertovať ich na ne
 (výnimka `InvalidCastException`):

(1,18): error CS1002: ; expected
(1,34): error CS1002: ; expected
(1,46): error CS1002: ; expected
(1,63): error CS1002: ; expected
(1,74): error CS1002: ; expected
(1,95): error CS1003: Syntax error, '(' expected
(1,112): error CS1003: Syntax error, ',' expected
(1,124): error CS1003: Syntax error, ',' expected
(1,135): error CS1003: Syntax error, ',' expected
(1,149): error CS1003: Syntax error, ',' expected
(1,157): error CS1001: Identifier expected
(1,157): error CS1003: Syntax error, ',' expected
(1,165): error CS1003: Syntax error, ',' expected
(1,167): error CS1001: Identifier expected
(1,184): error CS1003: Syntax error, ',' expected
(1,191): error CS1003: Syntax error, ',' expected
(1,217): error CS1003: Syntax error, ',' expected
(1,226): error CS1003: Syntax error, ',' expected
(1,231): error CS1001: Identifier expected
(1,231): error CS1003: Syntax error, ',' expected
(1,240): error CS8124: Tuple must contain at least two elements.
(1,240): error CS1026:) expected
(1,240): error CS1001: Identifier expected
(1,240): error CS1056: Unexpected character ``'
(1,241): error CS1003: Syntax error, ',' expected
(1,261): error CS1001: Identifier expected
(1,261): error CS1056: Unexpected character ``'
(1,264): error CS1018: Keyword 'this' or 'base' expected
(1,264): error CS1002: ; expected

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```
var numbers = new ArrayList();  
numbers.Add(1);  
numbers.Add(2);  
numbers.Add(3);  
numbers.Add("4");  
numbers.Add(3);  
  
foreach (int number in numbers)  
{
```

```
    Console.WriteLine(number);  
}
```

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System.InvalidCastException: Unable to cast object of type 'System.String' to type 'System.Int32'.

at Submission#24.<<Initialize>>d__0.MoveNext()

--- End of stack trace from previous location ---

at

Microsoft.CodeAnalysis.Scripting.ScriptExecutionState.RunSubmissionsAsync[TResult]
(ImmutableArray`1 precedingExecutors, Func`2 currentExecutor, StrongBox`1
exceptionHolderOpt, Func`2 catchExceptionOpt, CancellationToken
cancellationToken)

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Pri generickej verzii dostaneme kompilačnú chybu ('Error CS1503'), takže ju môžeme

(1,16): error CS1002: ; expected
(1,33): error CS1002: ; expected
(1,51): error CS1056: Unexpected character ``'
(1,64): error CS1056: Unexpected character ``'
(1,66): error CS1002: ; expected
(1,66): error CS7017: Member definition, statement, or end-of-file expected
(1,77): error CS1002: ; expected

(1,89): error CS1002: ; expected
(1,98): error CS1733: Expected expression

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```
var numbers = new List<int>();  
numbers.Add(1);  
numbers.Add(2);  
numbers.Add(3);  
numbers.Add("4"); // Error CS1503: Argument 1: cannot convert from 'string' to  
numbers.Add(3);
```

(5,13): error CS1503: Argument 1: cannot convert from 'string' to 'int'

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```
## Znovupoužitie binárneho kódu
```

(1,1): error CS1024: Preprocessor directive expected

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Výhoda generík je, že ich môžeme definovať raz a potom ich použiť s viacerými. Generiká môžu byť definované v jednom jazyku a potom použité v akomkoľvek inom. Príklad použitia triedy `List` s rôznymi typmi:

(1,16): error CS1002: ; expected
(1,18): error CS1002: ; expected
(1,18): error CS7017: Member definition, statement, or end-of-file expected
(1,27): error CS1002: ; expected
(1,44): error CS1002: ; expected
(1,50): error CS1002: ; expected
(1,60): error CS1002: ; expected
(1,69): error CS1002: ; expected

(2,10): error CS1003: Syntax error, '(' expected
(2,19): error CS1003: Syntax error, ',' expected
(2,32): error CS1003: Syntax error, ',' expected
(2,46): error CS1003: Syntax error, ',' expected
(2,54): error CS1003: Syntax error, ',' expected
(2,64): error CS1003: Syntax error, ',' expected
(2,80): error CS1003: Syntax error, ',' expected
(2,81): error CS1003: Syntax error, ',' expected
(2,84): error CS1001: Identifier expected
(2,84): error CS1003: Syntax error, ',' expected
(2,85): error CS1003: Syntax error, ',' expected
(2,96): error CS1003: Syntax error, ',' expected
(2,97): error CS1003: Syntax error, ',' expected
(3,18): error CS1003: Syntax error, ',' expected
(3,25): error CS1001: Identifier expected
(3,25): error CS1056: Unexpected character ``'
(3,26): error CS1003: Syntax error, ',' expected
(3,30): error CS1001: Identifier expected
(3,30): error CS1056: Unexpected character ``'
(3,32): error CS1003: Syntax error, ',' expected
(3,42): error CS1003: Syntax error, ',' expected
(3,47): error CS1001: Identifier expected
(3,47): error CS1003: Syntax error, ',' expected
(3,48): error CS1026:) expected
(3,48): error CS1002: ; expected

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```
var numbers = new List<int>();
numbers.Add(10);

var realNumbers = new List<double>();
realNumbers.Add(Math.PI);
realNumbers.Add(10);
realNumbers.Add(1.123);

var words = new List<string>();
words.Add("Hello");
words.Add("generics");

class Person
{
    public string FullName {get; set;}
}
```

```

    public DateTime Birthdate { get; set; }

    public Person(string fullName, DateTime birthdate) => (FullName, Birthdate)
}

var people = new List<Person>();
people.Add(new Person("Jan Mrkvicka", DateTime.Today));
people.Add(new Person("Elena Mrkvickova", DateTime.Now.AddYears(-20)));
foreach(var p in people)
    Console.WriteLine($"{p.FullName} ({p.Birthdate})");

```

Jan Mrkvicka (10/5/2026 12:00:00 AM)

Elena Mrkvickova (10/5/2006 10:23:26 PM)

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V IL kóde je definícia generickej triedy iba raz. Až JIT kompilátor (CLR) kompiluje. Pri hodnotových typoch JIT kompilátor vytvára novú triedu pre konkrétny použitý

```

(1,6): error CS1002: ; expected
(1,14): error CS1002: ; expected
(1,35): error CS1002: ; expected
(1,46): error CS1002: ; expected
(1,58): error CS1002: ; expected
(1,75): error CS1002: ; expected
(1,91): error CS1002: ; expected
(1,100): error CS1002: ; expected
(1,115): error CS1002: ; expected
(1,121): error CS1002: ; expected
(1,140): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement
(1,147): error CS1002: ; expected
(1,147): error CS1003: Syntax error, 'while' expected

```

(1,147): error CS1003: Syntax error, '(' expected
(1,150): error CS1026:) expected
(1,150): error CS1002: ; expected
(1,160): error CS1002: ; expected
(1,164): error CS1002: ; expected
(1,164): error CS7017: Member definition, statement, or end-of-file expected
(1,172): error CS1002: ; expected
(1,180): error CS1002: ; expected
(1,197): error CS1002: ; expected
(2,5): error CS1003: Syntax error, '(' expected
(2,24): error CS1003: Syntax error, ',' expected
(2,39): error CS1003: Syntax error, ',' expected
(2,52): error CS1003: Syntax error, ',' expected
(2,63): error CS1003: Syntax error, ',' expected
(2,81): error CS1003: Syntax error, ',' expected
(2,103): error CS1003: Syntax error, ',' expected
(2,113): error CS1003: Syntax error, ',' expected
(2,124): error CS1003: Syntax error, ',' expected
(2,129): error CS1001: Identifier expected
(2,140): error CS1003: Syntax error, ',' expected
(2,154): error CS1003: Syntax error, ',' expected
(2,159): error CS1001: Identifier expected
(2,172): error CS1003: Syntax error, ',' expected
(2,186): error CS1003: Syntax error, ',' expected
(2,207): error CS1003: Syntax error, ',' expected
(2,214): error CS1003: Syntax error, ',' expected
(2,227): error CS1056: Unexpected character 'â€‘'
(2,229): error CS1003: Syntax error, ',' expected
(2,239): error CS1001: Identifier expected
(2,239): error CS1003: Syntax error, ',' expected
(2,242): error CS1003: Syntax error, ',' expected
(2,248): error CS1001: Identifier expected
(2,248): error CS1003: Syntax error, ',' expected
(2,249): error CS1031: Type expected
(2,249): error CS8124: Tuple must contain at least two elements.
(2,249): error CS1026:) expected
(2,249): error CS1001: Identifier expected
(2,249): error CS1003: Syntax error, ',' expected
(2,251): error CS1003: Syntax error, ',' expected
(2,256): error CS1001: Identifier expected
(2,258): error CS1002: ; expected
(2,264): error CS1002: ; expected
(2,267): error CS1002: ; expected
(2,276): error CS1003: Syntax error, ',' expected
(2,284): error CS1002: ; expected

```
(2,296): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement
(2,299): error CS1002: ; expected
(2,308): error CS1002: ; expected
(2,318): error CS1002: ; expected
(2,324): error CS1002: ; expected
(2,334): error CS1002: ; expected
(2,347): error CS1002: ; expected
(2,347): error CS7017: Member definition, statement, or end-of-file expected
(2,365): error CS1002: ; expected
(2,374): error CS1002: ; expected
(2,389): error CS1002: ; expected
(2,394): error CS1002: ; expected
(2,394): error CS7017: Member definition, statement, or end-of-file expected
(2,405): error CS1002: ; expected
(2,419): error CS1002: ; expected
(2,437): error CS1001: Identifier expected
(2,437): error CS1003: Syntax error, '(' expected
(2,437): error CS1026: ) expected
(2,437): error CS1002: ; expected
```

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```
## Pomenovanie
```

```
(1,1): error CS1024: Preprocessor directive expected
```

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- Názov prefixu generického typu začína na písmeno T
- Ak generický typ môže byť nahradený akýmkoľvek typom a je iba jeden typ použ
- Ak je špeciálna požiadavka (nastavené obmedzenie) na generický typ alebo ak

```
(1,9): error CS1002: ; expected
(1,29): error CS1002: ; expected
(1,41): error CS1002: ; expected
(1,52): error CS1002: ; expected
(2,6): error CS1002: ; expected
```

(2,20): error CS1002: ; expected
(2,29): error CS1002: ; expected
(2,50): error CS1002: ; expected
(2,58): error CS1002: ; expected
(2,65): error CS1002: ; expected
(2,75): error CS1002: ; expected
(2,82): error CS1002: ; expected
(2,82): error CS7017: Member definition, statement, or end-of-file expected
(2,93): error CS1002: ; expected
(2,105): error CS1002: ; expected
(2,123): error CS1002: ; expected
(2,132): error CS1002: ; expected
(3,6): error CS1002: ; expected
(3,19): error CS1002: ; expected
(3,41): error CS1003: Syntax error, ',' expected
(3,53): error CS1002: ; expected
(3,66): error CS1002: ; expected
(3,76): error CS1002: ; expected
(3,82): error CS1002: ; expected
(3,94): error CS1002: ; expected
(3,101): error CS1002: ; expected
(3,106): error CS1002: ; expected
(3,106): error CS7017: Member definition, statement, or end-of-file expected
(3,119): error CS1002: ; expected
(3,133): error CS1002: ; expected
(3,147): error CS1002: ; expected
(3,169): error CS1001: Identifier expected
(3,177): error CS1001: Identifier expected
(3,183): error CS1001: Identifier expected
(3,191): error CS1001: Identifier expected
(3,193): error CS1031: Type expected
(3,193): error CS1001: Identifier expected
(3,193): error CS1056: Unexpected character 'â€'|'
(3,195): error CS1002: ; expected

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```
## Generické metody
```

(1,1): error CS1024: Preprocessor directive expected

```
static void ProcessItems<T>(IList<T> list)
{
    System.Console.WriteLine($"IsReadOnly vracia {list.IsReadOnly} pre {list.GetType().Name}");

    //coll.RemoveAt(4);

    foreach (T item in list)
    {
        System.Console.Write(item.ToString() + " ");
    }
    System.Console.WriteLine();
}

int[] arr = { 0, 1, 2, 3, 4 };
List<int> list = new List<int>();

for (int x = 5; x < 10; x++)
{
    list.Add(x);
}

ProcessItems<int>(arr);
ProcessItems<int>(list);
```

IsReadOnly vracia True pre System.Int32[].

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IsReadOnly vracia False pre System.Collections.Generic.List`1[System.Int32].

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